HPC AI
How HPC Supports AI Research for Thai Stocks

5

This project builds a controlled, verifiable process to compare five computer-learning methods in a Thai-stock setting.

503

planned experiment runs

5

AI

learning algorithms

29.7%

verified official-plan progress

HPC

· HPC makes the project manageable, traceable and repeatable.

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What Is the Computer Learning?

Deep reinforcement learning repeats decisions and learns from feedback; the five names are different learning styles.

PPO | **A2C** | **DDPG** | **TD3** | **SAC**

1 · / Observe

Read prices, trading volume and market context.

2 · / Decide

Choose how much of the portfolio goes to each stock.

3 · / Learn

Learn from returns, risk and transaction costs.

4 · / Test

Evaluate on later dates not used for training.

· Repeated evidence, not one lucky result.

HPC ?
Why Does HPC Matter?

503

A laptop is useful for small tests, but a 503-run plan must be scheduled, recorded, checked and reproducible.

PERSONAL COMPUTER

- / build and debug code
- / run small trials
- / inspect results quickly

HPC
HIGH-PERFORMANCE COMPUTING

- / schedule tasks
- / record job IDs, runtime and resources
- / verify, recover and repeat

HPC **· HPC supports speed and research reliability.**

A Real HPC Run

/ STORAGE-SAFE SMOKE TEST

```
#                                          1
job_id=$(sbatch --parsable smoke_test.sbatch)

#
Job 3550
State: COMPLETED
Elapsed: 00:33:20
Algorithms: 5
```

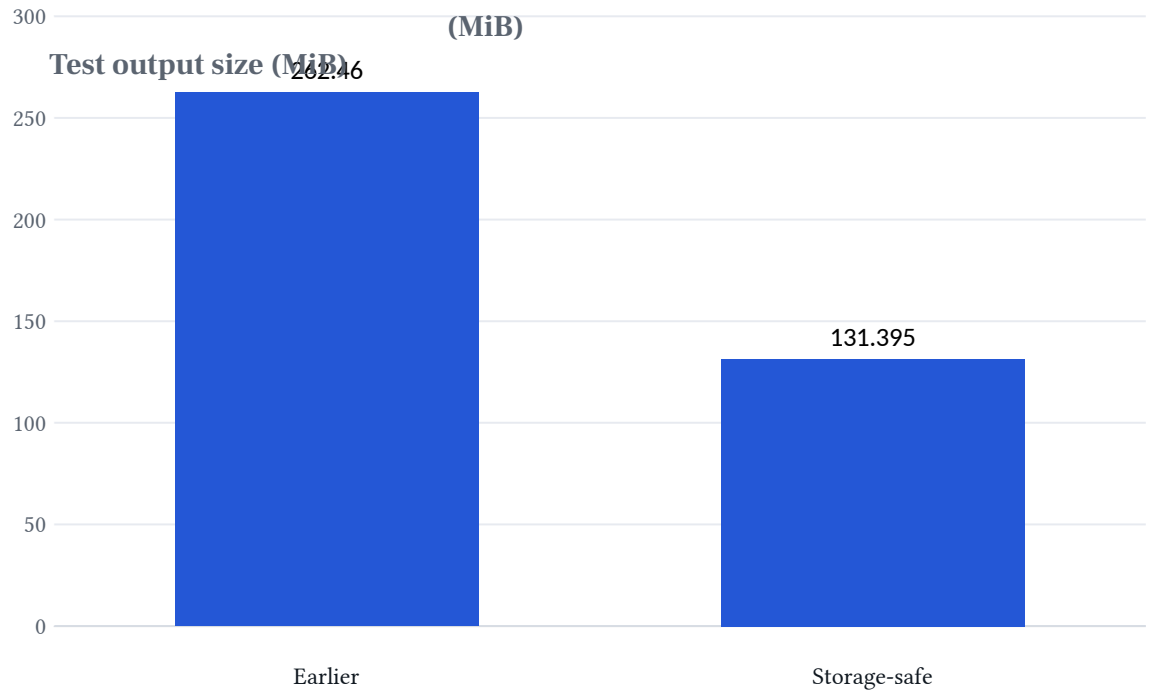
Smoke test ?
What is a smoke test?

A small trial that checks the complete process before a large experiment is submitted.

/ COMPLETED
33 20 / 33 min 20 sec

· Large campaigns begin with a checkable trial.

Responsible Use of Storage



MiB / MiB is a standard file-size unit.

49.94% / less output

5 ✦ 5 5 + 5

5 manifests + 5 retention receipts

0 large intermediate checkpoints

· Output was nearly halved while verification evidence was preserved.

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What Is Done and What Remains?

/ VERIFIED

11 / 37

(29.7%)

official project checkpoints complete

- HPC
-

- Storage-safe HPC test completed
- All five methods produced complete records

/ STILL AHEAD

150
150 trials

full screening has not been submitted

:
:

- Remaining checkpoints are incomplete
- No final research conclusion yet

“ ” “ ” · **Ready is not the same as finished.**

What Happens Next?

1

Recheck capacity

Confirm storage and compute before a new window.

2

150

Run 150-trial screening

Submit only when fresh safety checks pass.

3

Compare and preserve

Report all five methods with reproducible records.

HPC

HPC turns a large idea into research that can be measured, checked and repeated.